

JULIAN T. VANCE

Senior Director of Cyber-Physical Security & Infrastructure Resilience

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PROFESSIONAL SUMMARY

Technically elite and strategically driven Cyber-Physical Security Executive with over 12 years of specialized experience safeguarding global critical infrastructure, renewable energy grids, and large-scale industrial automation systems. Recognized expert in the convergence of Information Technology (IT) and Operational Technology (OT), with a definitive record of defending multi-billion dollar asset portfolios against advanced persistent threats (APTs) and sophisticated nation-state actors. Distinguished for architecting the "Resilient Grid 2030" framework, which integrated cutting-edge zero-trust architecture into legacy SCADA/ICS environments for the largest utility providers in the ERCOT region.

Julian has a proven ability to lead high-stakes, large-scale incident response missions in sensitive air-gapped environments. He has successfully managed annual security budgets exceeding \$180M and directed cross-functional teams of 120+ cyber-physical engineers. He possesses deep expertise in navigating complex regulatory landscapes, including NERC CIP, NIST SP 800-82, and IEC 62443, ensuring 100% compliance across 45+ high-voltage substations and 12 major generation facilities. As a strategic advisor to the Department of Energy (DOE) on grid modernization, he is a frequent contributor to international standards on industrial protocol security and critical infrastructure protection. He is highly regarded for his unique ability to translate high-level cyber risk into operational reality for C-Suite executives and Board Directors, fostering a culture of "Safety-First Security."

CORE COMPETENCIES

- **Cyber-Physical Security Strategy:** OT/ICS Risk Management, Grid Modernization, SCADA Security, Industrial Zero Trust, Threat Landscape Analysis.
- **Operational Excellence:** NERC CIP Compliance, NIST Cybersecurity Framework (CSF), ISA/IEC 62443 Standards, Threat Modeling, Security Orchestration.
- **Incident Response & Resilience:** Cyber-Physical Forensic Analysis, Disaster Recovery for ICS, Business Continuity, Root Cause Analysis, Post-Breach Remediation.
- **Infrastructure Leadership:** Multi-Site Security Operations (SecOps), Strategic Asset Protection, CapEx/OpEx Budgeting (\$100M+), Executive Stakeholder Management.
- **Technical Innovation:** Software Defined Networking (SDN) for OT, ICS-Specific IDS/IPS Deployment, Blockchain for Grid Integrity, Edge Computing Security.

- **Regulatory Navigation:** FERC/NERC Audit Management, Federal Information Security Modernization Act (FISMA), Supply Chain Risk Management (SCRM).
- **Advanced Threat Defense:** APT Hunting in OT Networks, Firmware Security Analysis, Protocol Deep Packet Inspection (DPI), Honeypot Deployment in Industrial Webs.

TECHNICAL SKILLS

Industrial Control Systems (ICS) & SCADA Architecture

- **PLCs & RTUs:** Expert-level configuration and security hardening for Siemens S7-1500, Allen-Bradley ControlLogix, Schneider Electric Modicon, Emerson DeltaV, Honeywell Experion, and Yokogawa Centum VP.
- **Industrial Protocols:** Mastery of Modbus TCP/RTU, DNP3, IEC 60870-5-104, BACnet, Profinet, OPC UA, EtherNet/IP, GOOSE (IEC 61850), and S7Comm.
- **Human Machine Interfaces (HMI):** Advanced deployment of Wonderware InTouch, Ignition by Inductive Automation, GE Digital iFIX, Rockwell FactoryTalk, and VTScada.

Security Platforms & Visibility Tools

- **OT Visibility & Threat Detection:** Dragos Platform (Expert), Nozomi Networks Guardian, Claroty Continuous Threat Detection (CTD), Microsoft Defender for IoT (CyberX), and Armis.
- **Industrial Network Security:** Palo Alto Networks (Industrial Firewalls & Strata), Fortinet FortiGate Rugged, Waterfall Unidirectional Gateways, and Owl Cyber Defense Data Diodes.
- **SIEM, SOAR, & XDR:** Splunk Enterprise Security (OT Add-on), Exabeam, IBM QRadar, Palo Alto Cortex XSOAR, and Phantom.

Systems, Cloud, & Network Architecture

- **Virtualization & Containerization:** VMware vSphere (ESXi), Microsoft Hyper-V, Docker, Kubernetes for Edge Gateway Management, and Proxmox.
- **Hardened Operating Systems:** Windows Server (OT Hardened), RHEL (Industrial), QNX, VxWorks, FreeRTOS, and various embedded Linux distributions.
- **Digital Forensics:** SANS SIFT Workstation, Mandiant Redline, EnCase Forensic, Wireshark (Custom Dissectors for Proprietary Industrial Protocols), and Volatility.

Development, Automation, & Orchestration

- **Programming Languages:** Python (Security Tooling & Scripting), C++ (Embedded Security), PowerShell, Bash, Structured Text (IEC 61131-3).
- **Security Automation:** Ansible for OT Hardening, Terraform for Infrastructure-as-Code (IaC), Jenkins for Secure CI/CD in Edge Computing, and SaltStack for configuration management.

PROFESSIONAL EXPERIENCE

Senior Director of Infrastructure Cybersecurity | Sentinel Grid Solutions | 2021 – Present

Houston, TX (Global Leader in Grid Automation & Critical Infrastructure | Revenue: \$12B)

Strategic Leadership & Operational Mandate: Julian was recruited to lead the global Cyber-Physical Security Division, where he oversees the defense of a critical client asset portfolio that includes 18% of the U.S. electrical transmission grid and 22 major international hydroelectric facilities. He manages an annual budget of \$185M and leads a diverse, high-performing workforce of 140 engineers, forensic analysts, and physical security specialists.

Key Strategic Achievements:

- **Grid Modernization Initiative (Project Sentinel Shield):** Led a \$400M, multi-year digital transformation program that replaced 15,000 legacy serial-based RTUs with secure-by-design IEC 62443 certified components across 250 substations. This initiative reduced the overall attack surface by 68% and improved operational uptime by 12% through proactive health monitoring.
- **Critical Incident Response:** Directed the remediation of a sophisticated nation-state supply chain breach that targeted firmware in localized grid controllers. Mobilized an 80-person rapid response team that successfully identified, tested, and patched 12,000 units in 14 days without a single localized blackout, preventing an estimated \$1.2B in potential economic disruption.
- **OT-SOC Establishment:** Built and operationalized the first industry-specific OT Security Operations Center (SOC). This SOC utilizes custom AI-driven anomaly detection models to monitor 4.5 million telemetry points per second. Since its launch, Julian has seen threat detection speeds improve by 400% and false positive rates drop by 85%.
- **Regulatory Compliance & Governance:** Achieved "Zero Non-Compliance" status across 14 simultaneous NERC CIP Version 7 audits. He developed a proprietary "Compliance Automation Engine" (CAE) that reduced audit preparation time by 1,200 man-hours annually by automating evidence collection from disconnected ICS assets.
- **Executive Advising & Board Presence:** Serves as the primary security liaison to the Board of Directors. Julian successfully secured an additional \$50M in funding for "Cyber-Resilience-as-a-Service" (CRaaS) aimed at municipal utility clients who lack specialized OT security staff.
- **Strategic M&A Operations:** Led the cybersecurity due diligence for the \$2.4B acquisition of "Vortex Power Systems." He identified \$15M in critical security debt and successfully integrated their vast OT assets into the Sentinel core framework within 120 days.

Director of OT Security & Automation | Meridian Energy Systems | 2017 – 2021

Dallas, TX (Integrated Energy Utility | Annual Revenue: \$7.5B)

Operational Mandate: Julian directed the security operations for a sprawling energy portfolio comprising thermal, wind, and solar generation. He was responsible for the lifecycle security of Distributed Energy Resources (DERs) and the primary SCADA control center for the entire Southwest region.

Key Measurable Achievements:

- **Zero Trust for ICS Architecture:** Architected and deployed the first full-scale Zero Trust Network Access (ZTNA) model for a thermal power plant environment. By eliminating all VPN-based lateral movement risks, he reduced unauthorized access attempts and internal reconnaissance by 99.7%.
- **Supply Chain Security Program:** Established a "Trusted Component Vendor" (TCV) program that mandated hardware-root-of-trust for all new equipment. Julian's team discovered and blocked 4 critical vulnerabilities in pre-release firmware from a Tier-1 vendor, resulting in a global product recall and saving the firm from massive liability.
- **Asset Visibility Project (Operation Lens):** Deployed a passive monitoring solution (Nozomi Guardian) across 100% of the generation fleet. The project discovered 15% more "ghost assets" than previously documented in manual inventory, leading to a comprehensive hardening of the network perimeter.
- **Cyber-Physical Red Teaming:** Designed and executed the industry's most realistic "Grid-Down" simulation in a controlled lab environment. This project identified 12 critical logic flaws in the automatic load-shedding sequence that could have caused catastrophic equipment damage during a real cyber-physical event.
- **Operational Cost Reduction:** Automated the vulnerability management process for OT assets using customized Ansible playbooks. This reduced the cost-per-remediation from \$450 to \$180 and cut the window from disclosure to patch-testing by 60%.
- **Public-Private Partnership:** Acted as the Meridian lead for the Department of Energy's "Project CyTrus," sharing real-time threat intelligence with the Electricity Information Sharing and Analysis Center (E-ISAC).

Senior Cyber-Physical Security Engineer | Ironwood Automation | 2014 – 2017

Chicago, IL (Industrial Automation & Systems Integrator)

Strategic Mandate & Engineering Leadership: Julian acted as the lead security architect for large-scale automation projects in the petrochemical and pharmaceutical manufacturing sectors. He was pivotal in bridging the gap between mechanical engineering teams and IT security departments.

Key Measurable Achievements:

- **Petrochemical Plant Hardening:** Lead the security design for a \$1.5B refinery expansion for fictional client "Crestwood Refining." He implemented unidirectional gateways (data diodes) to protect the Safety Instrumented Systems (SIS), ensuring physical safety even during a complete network compromise.
- **PLC Hardening Framework (The Ironwood Standard):** Authored a 200-page configuration manual for PLC hardening, which became the mandatory standard for all subsequent systems integration projects at the company. This standard reduced post-deployment security tickets by 55% across all clients.
- **Training & Workforce Development:** Created a "Cyber-Aware Operator" curriculum specifically for plant floor workers. He trained 1,200+ employees on recognizing physical signs of cyber interference, which led to the early detection of two localized system malfunctions that were subsequently identified as unauthorized configuration changes.
- **Secure Remote Access Re-Architecture:** Developed a hardened jump-host architecture for third-party maintenance contractors. By implementing multi-factor authentication (MFA) and full session recording in a protocol-isolated environment, he eliminated the most common vector for industrial malware entry.
- **Advanced Network Segmentation:** Redesigned the Purdue Model implementation for a major pharmaceutical manufacturer, separating the manufacturing execution system (MES) from the enterprise ERP using high-availability industrial firewalls and custom-tailored ACLs.

Cyber-Physical Systems Analyst | CoreStream Power | 2011 – 2014

Charlotte, NC (Transmission & Distribution Specialist)

Early Career Foundations: Julian focused on the technical analysis of industrial protocols and the physical security of remote substation assets. He was one of the first analysts in the company to apply IT security principles to legacy serial communication devices.

Key Measurable Achievements:

- **DNP3 Protocol Deep-Dive:** Developed a custom Wireshark dissector to identify malformed DNP3 packets used in potential denial-of-service attacks against older RTUs. This tool was later shared with the open-source community.
- **Substation Security Audits:** Personally conducted 45 physical and logical security audits across the Eastern Interconnect. He identified and remediated 120+ critical findings related to exposed serial-to-ethernet converters and insecure cabinet locks.
- **Baseline Configuration Management:** Implemented a Tripwire-based integrity monitoring system for Windows-based HMI stations. This initiative reduced unauthorized configuration changes by 90% and provided the first reliable audit trail for operator actions.
- **NERC CIP v3 to v5 Transition Taskforce:** Acted as a key member of the compliance task force during the massive NERC CIP v5 overhaul. He mapped technical controls to the new regulatory requirements for over 500 assets, ensuring a seamless transition without fines or warnings.

KEY PROJECTS PORTFOLIO

Project Aegis: The Resilient Grid 2030 Transformation

Role: Executive Project Sponsor & Lead Architect

Scope: \$400M / 36 Months / 250 Substations

- **Objective:** Modernize the legacy transmission infrastructure of a regional utility to support high-penetration renewable energy while hardening against nation-state cyber attacks.
- **Strategy:** Julian deployed a hybrid Software Defined Network (SDN) for the OT backplane, allowing for micro-segmentation of GOOSE and Sampled Value traffic. He integrated hardware-accelerated encryption for all inter-substation communications.
- **Outcome:** Achieved a "Self-Healing" network capability that can isolate a compromised substation from the rest of the grid in less than 50ms. The project was cited as a "Global Best Practice" by the North American Electric Reliability Corporation (NERC).

Operation Silver-Thread: Zero Trust Transition

Role: Project Director

Scope: \$85M / 18 Months / 12 Generation Facilities

- **Objective:** Move away from traditional "Crunchy Perimeter / Soft Interior" security models to a granular Zero Trust architecture for Power Generation assets.
- **Strategy:** Implemented Identity-Aware Proxies (IAP) at the plant level and migrated all SCADA traffic to an encrypted overlay network. Julian used behavioral analytics to establish "Normal" baselines for every PLC/HMI interaction.
- **Outcome:** Prevented a lateral movement attempt during a multi-week Red Team exercise where the "attacker" had gained physical access to a site. Zero unauthorized commands were successful during the test.

The "Black-Start" Resilience Framework

Role: Technical Lead

Objective: Ensure the ability to restart the electrical grid from a total blackout scenario even in the presence of an ongoing, active cyber attack.

- **Strategy:** Developed a set of "Out-of-Band" (OOB) immutable backup systems for critical generation settings. Julian created a satellite-linked communication bridge that bypasses compromised fiber networks.
- **Outcome:** Reduced theoretical "Black-Start" time from 48 hours to 12 hours under contested cyber conditions. This framework has been adopted by three regional ISOs as part of their emergency planning.

LEADERSHIP & TEAM MANAGEMENT EXPERIENCE

- **Executive Recruitment:** Julian personally built the "Cyber-Physical Intelligence Team" (CPIT) at Sentinel, recruiting 40 top-tier experts from the NSA, DOE, and private sector industrial labs to create a world-class threat hunting unit.
- **Crisis Management Framework:** Developed a standardized "Security Incident & Disaster Recovery" (SIDR) playbook used by 12 major utilities. He has conducted over 50 "Tabletop-to-Real-Life" drills involving C-level executives, state governors, and emergency management agencies.
- **Mentorship & Talent Pipeline:** Launched a "Bridging the Gap" initiative that pairs veteran Mechanical Engineers with junior Cyber Analysts. This program resulted in a 30% increase in cross-functional skill sets within its first year and significantly reduced friction between "IT" and "OT" departments.
- **Global Team Scaling:** Successfully managed a remote, distributed workforce during the COVID-19 pandemic, ensuring 24/7/365 security coverage of the grid while maintaining a 95% employee retention rate.
- **Budgetary Stewardship:** Consistently delivered projects under budget by leveraging open-source components for non-critical monitoring and focusing capital on high-impact proprietary defense systems.

PROCESS IMPROVEMENTS & AUTOMATION ACHIEVEMENTS

- **Automated Vulnerability Mapping (V-Map):** Julian built a proprietary tool that automatically maps NIST National Vulnerability Database (NVD) entries to the specific firmware versions found in the field via Nozomi/Dracos API. This reduced vulnerability triage time from weeks to minutes.
- **Hardening-as-Code (HaC):** Implemented a library of hardened Ansible roles for various industrial OS flavors (QNX, VxWorks, Windows IoT). He ensured that every new device deployed to the field meets 100% of the security baseline before it is ever energized.
- **Digital Twin Simulation Lab:** Established a "Security Digital Twin" lab that mimics the exact network topology and device configuration of every facility in the fleet. This allows for the risk-free testing of patches and configuration changes on a virtualized mirror before deployment to live production systems.
- **Supply Chain Risk Automation:** Developed a "Supplier Risk Portal" that uses machine learning to analyze the security posture of third-party vendors based on public data leaks, financial stability, and historical security performance.

EDUCATION

Master of Science in Cybersecurity Policy & Grid Engineering

Carnegie Mellon University, Pittsburgh, PA

- Focus Areas: Security of Cyber-Physical Systems, Cryptography for Embedded Systems, Policy Frameworks for Critical Infrastructure.
- Thesis: "Mitigating Cascading Failures in Electrical Grids via Distributed Anomaly Detection."
- Honors: CMU Excellence in Engineering Research Award.

Bachelor of Science in Electrical Engineering & Computer Science (EECS)

University of California, Berkeley, CA

- Concentration: Control Systems, Power Electronics, Network Security.
- Honors: Eta Kappa Nu (HKN) Electrical Engineering Honor Society; Dean's List for 7 consecutive semesters.

CERTIFICATIONS

- **GICSP:** Global Industrial Cyber Security Professional – GIAC
- **GRID:** GIAC Response and Industrial Defense – GIAC
- **CISSP-ISSAP:** Information Systems Security Architecture Professional
- **CISM:** Certified Information Security Manager – ISACA
- **PMP:** Project Management Professional – PMI
- **ISA/IEC 62443 Cybersecurity Expert Certification**
- **SCADA Security Architect (CSSA)** – IACRB
- **CEH:** Certified Ethical Hacker (Industrial Focus)

TOOLS & TECHNOLOGIES REFERENCE

Industrial Visibility & Threat Detection:

- Dragos Platform, Nozomi Guardian, Claroty Continuous Threat Detection (CTD), Armis, Microsoft Defender for IoT, Sentryo.

Networking & Boundary Protection:

- Palo Alto Networks PA-Series (Rugged), Fortinet FortiGate, Cisco ISA 3000, Tofino Xenon Security Appliance, Waterfall Unidirectional Gateways, Owl Cyber Defense Data Diodes, Hirschmann Industrial Switches.

Security Operations & Forensics:

- Splunk ES, Exabeam, IBM QRadar, SANS SIFT, Wireshark, Volatility, FTK Imager, Autopsy, Mandiant Redline.

Engineering & Automation:

- TIA Portal (Siemens), RSLogix 5000 (Rockwell), MATLAB/Simulink, Ansible, Terraform, Git, Docker, Python, C++, SQL, PowerBI.

PUBLICATIONS & CONFERENCES

Selected Peer-Reviewed Publications

- **"Zero Trust in Air-Gapped Environments: A Paradoxical Necessity."**
IEEE Security & Privacy Magazine, Vol. 19, Issue 2, 2023.
- **"Analyzing the Vulnerabilities of DNP3 in the Context of Nation-State Attacks."**
Journal of Critical Infrastructure Protection, 2021.
- **"Software Defined Networking for IEC 61850: A New Frontier for Grid Resilience."**
International Journal of Power & Energy Systems, 2019.

Keynote & Speaking Engagements

- **Keynote:** "Defending the ERCOT Grid: Lessons from the Digital Frontline," *S4x24 (SCADA Security Scientific Symposium)*, Miami, 2024.
- **Featured Speaker:** "The Convergence of Physical and Cyber Safety in High-Pressure Environments," *Black Hat USA (ICS Track)*, Las Vegas, 2022.
- **Panelist:** "Global Grid Security: A Bipartisan Approach to Resilience," *World Energy Congress*, 2021.
- **Workshop Lead:** "Advanced ICS Malware Analysis," *DEF CON (ICS Village)*, 2020.

PROFESSIONAL AFFILIATIONS

- **Fellow:** The Energy Infrastructure Security (EIS) Council.
- **Senior Member:** IEEE Power & Energy Society (PES).
- **Advisory Board Member:** International Society of Automation (ISA) – Security Standards Committee.
- **Technical Consultant:** NERC CIP Standards Drafting Team (Technical Consultant for v7).
- **Executive Mentor:** SANS Women in Cybersecurity – Promoting diverse talent in the ICS field.
- **Member:** InfraGard (Houston Chapter).

STRATEGIC VISION: THE FUTURE OF CYBER-PHYSICAL RESILIENCE

The Paradigm Shift

In the coming decade, the traditional boundaries between "Digital Security" and "Physical Safety" will vanish entirely. As we move toward a decentralized, 100% renewable grid populated by millions of smart edge devices, the complexity of our systems will increase by orders of magnitude. My vision is the creation of an **Immutable Infrastructure**—a grid that is not just "hardened" but "resilient by design."

Core Values of My Leadership

- **Security as a Safety Metric:** In the industrial world, a cyber-breach is a safety event. My teams prioritize human life and environmental protection above all digital metrics. We treat a malformed packet as seriously as a high-pressure valve failure.
- **Radical Transparency with Vendors:** We do not accept "Security through Obscurity." Julian mandates that all vendors provide Software Bills of Materials (SBOMs) and open themselves to third-party hardware audits as a condition of procurement.
- **Continuous Learning:** The threat landscape evolves faster than any three-year procurement cycle. We build systems that are flexible enough to be updated in real-time without taking the grid offline.

Final Commitment

My career has been dedicated to a singular, unwavering goal: ensuring that the lights stay on, the water stays clean, and the machines keep running—no matter who is trying to stop them.